FNCE 30007 Derivative Securities

Tutorial 6 Questions

The Binomial Model (Part II)

QUESTION 1

Consider a 3-month American put option on a non dividend paying stock when the stock price is \$50, the strike price is \$50, the risk-free rate of interest is 12% p.a. and the underlying stock price is expected to increase or decrease by 2% per month.

(i) Use a three-period model to calculate the current value of the option.

(ii) Should the option be exercised early? Check your answer using the simulator.

QUESTION 2

Consider a two-period binomial model. Let $p = \dfrac{e^{r\Delta t} - d}{u - d}$ be the probability of a stock price increase in a risk neutral world. Use the principle of Risk Neutral valuation to derive the two period (European) valuation formula:

$$f = \frac{p^2 f_{uu} + 2p(1-p) f_{ud} + (1-p)^2 f_{dd}}{e^{2r\Delta t}}$$

Options on Futures

QUESTION 3

Using two portfolios, derive the lower bound for the price of a European futures put option. In doing this, detail the composition of each portfolio and draw a table illustrating the initial and terminal values of the portfolios.

QUESTION 4

Suppose you buy a put option contract on October gold futures with a strike price of \$900 per ounce. Each contract is for the delivery of 100 ounces. What happens if you exercise when the October futures price is \$880?

QUESTION 5

Suppose that a futures price is currently 30. The risk-free interest rate is 5% per annum. A three-month American call futures option with a strike price of 28 is worth 4. Calculate bounds for the price of a three-month American put futures option with a strike price of 28.

QUESTION 6

After reading the Virgin case, complete the following questions before the tutorial.

All market data are as at 2 September 2026.

	Option A	Option B	Option C
Underlying futures contract	Nov 2026 Brent	Dec 2026 Brent	Jan 2027 Brent
Futures price, F_0	95.63	92.09	88.98
Call strike, K	95.00	92.00	89.00
Option expiry	25 Sep 2026	27 Oct 2026	25 Nov 2026
Days to expiry	23	54	83
Implied volatility, σ	45.94%	46.88%	45.57%
Observed option price	4.67	6.70	7.72

Assume a continuously compounded risk-free rate of: r = 3.87% p.a.

Now focus on Option C: Jan 2027 Brent Call, which has the longest time to maturity. Revalue the option using: N = 2, 4, 8, 16

2.1 For each value of N, recalculate $\Delta t = T/N$ and the corresponding u, d and p. Use the simulator to construct the larger binomial trees where appropriate and compare the estimates in this table.

Number of Periods, N	Binomial Value	Observed Market Price	Difference: Model − Market	Absolute % Pricing Error
2		7.72		
4		7.72		
8		7.72		
16		7.72		

2.2 Interpretation

- How does the estimated option value change as the number of periods increases?
- Does the binomial estimate appear to become more stable as the tree becomes finer?
- How does the pricing error change as N increases?
- Does increasing the number of periods necessarily make the model value exactly equal to the observed market price? Briefly explain.

Data note: Option-chain figures are based on Barchart market data observed on 2 September 2026. Prices are quoted in USD per barrel. Barchart reports an option point value of USD 1,000.